

PROBLEM STATEMENT

Executive Summary

An organization with 1,480 employees is experiencing critical attrition challenges, losing 241 employees (16.28% attrition rate) annually. Analysis reveals that overtime-working employees face a 31.19% attrition rate—nearly three times higher than non-overtime workers. The problem necessitates immediate, data-driven intervention to understand root causes and implement retention strategies.

Problem Definition

Primary Problem: High employee attrition across the organization, particularly concentrated in specific departments and among overtime workers.

Secondary Issues:

- Disproportionate attrition in critical departments (HR: 26.83%, Sales: 25.23%)
- Strong correlation between overtime and attrition (31.19% vs 10.38%)
- Loss of talent despite competitive compensation
- Impact on organizational capability and growth objectives

Data Context

Dataset Size: 1,480 employees with 37 features including demographics, job characteristics, compensation, satisfaction metrics, and attrition status.

Key Metrics:

- Total Attrition: 241 employees (16.28%)
- Retained: 1,239 employees (83.72%)
- Age Range: 19-60 years (avg 36.9)
- Departments: 7 (Operations, Administration, Sales, IT, Marketing, HR, Finance)
- Salary Range: \$1,055 - \$99,806 per month

Key Findings from Preliminary Analysis

Factor	Attrition Rate	Key Insight
Overtime: Yes	31.19%	Critical risk factor
Overtime: No	10.38%	Baseline rate
HR Department	26.83%	Highest department loss
Sales Department	25.23%	Second highest risk
Operations Dept	8.33%	Lowest attrition
Admin Department	20.71%	Above average risk

Research Questions

1. What are the primary drivers of employee attrition in this organization?
2. How do overtime requirements correlate with attrition, and why is the impact so significant?
3. Are there demographic patterns (age, gender, marital status) associated with attrition?
4. Which job roles and departments are most vulnerable to talent loss?
5. What is the relationship between job satisfaction metrics and attrition?
6. How do compensation and benefits relate to retention decisions?

Solution Scope

Scope: Develop a comprehensive HR Analytics Dashboard that provides:

- Real-time attrition monitoring and early warning systems
- Department-wise and role-wise attrition trend analysis
- Root cause analysis through satisfaction and demographic correlations
- Executive dashboards for strategic decision-making
- Actionable insights for HR intervention planning

Success Criteria

- ✓ Reduce organization-wide attrition from 16.28% to <10% within 12 months
- ✓ Reduce overtime-related attrition from 31.19% to <15% within 6 months
- ✓ Identify and mitigate top 5 attrition factors
- ✓ Enable HR to predict and proactively address flight risks
- ✓ Provide actionable insights for retention strategy development